



M. Tech.

ELECTRICAL ENGINEERING

ELECTRICAL POWER SYSTEM

Syllabus

AY: 2019-20



Authored By: Charusat



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ACADEMIC REGULATIONS & SYLLABUS

2019-20

(Choice Based Credit System)

Faculty of Technology & Engineering (FTE)
M & V Patel Department of Electrical Engineering
Chandubhai S. Patel Institute of Technology (CSPIT)



Master of Technology Programme
(Electrical Power System)
(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

PROGRAM EDUCATIONAL OBJECTIVES

The objectives of the Electrical Engineering Postgraduate Program:

1. Apply their knowledge and expertise to find solutions of the problems in core and allied areas of engineering, exhibiting creativity and entrepreneurial skills.
2. Conduct themselves in an ethical and professional manner, exhibiting effective skills in communication, management, teamwork and leadership.
3. Continue their intellectual development through self-study and continuing education, practicing life-long learning.

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems
PO 2	An ability to write and present a substantial technical report/document.
PO 3	Students would be able to work individually and in the team for betterment of the society.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Can provide innovative solutions to advanced issues of power industry and society with the use of advanced technologies, modern tools and analytical skills.
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Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Master of Technology (Electrical Power System) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Academic Year 2019-2020

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Master of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

The Semester system of education should be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. *Duration of Programme*

Postgraduate programme	(M. Tech.)
Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

3. *Eligibility for admissions*

Minimum second class is required for admission into M.Tech programme.

4. *Mode of admissions*

Admission to M.Tech. programme will be as per Government of Gujarat guidelines. The eligibility norms require a condition to have a bachelor degree in related field and marks obtained in qualifying exam (like GATE) or common entrance test of Government of Gujarat. The detail eligibility norms will be as per Government of Gujarat guidelines.

5. *Programme structure and Credits*

As per annexure – I attached

6. *Attendance*

All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal. Student's attendance in a course should be 80%.

7 *Course Evaluation*

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.1 Final examination by the University through modes such as; written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, is set to 70% of the marks for each the course.

7.2 *Internal Evaluation*

- 7.2.1 A student shall be evaluated through Continuous Evaluation and Semester End Examination.
- 7.2.2 The weight of continuous assessment and End-semester examination shall be varying from UG to PG and from Faculty to Faculty as approved by Academic Council.
- 7.2.3 During the semester, a student shall be going through continuous assessment. The continuous assessment will be conducted by the respective Department / Institute. At the end of semester a student shall be evaluated through semester end examination comprising of theory and/or practical, viva-voce, term work components as decided by Academic Council.
- 7.2.4 The performance of candidate in continuous assessment and in end-semester examination together shall be considered for deciding the final grade in a course.

7.3 *University Examination*

- 7.3.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
- 7.3.2 In order to earn the credit in a course a student has to obtain grade other than FF.

7.3 Performance at Internal & University Examination

- 7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows:

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	50%

- 7.3.2 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8. Grading

- 8.1. The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme.

Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	09	08	07	06	05	04	00

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
And $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.

9. Award of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He/She should have earned minimum required credits as prescribed in course structure; and
 - 9.1.2 He/She should have cleared all internal and external evaluation components in every course; and
 - 9.1.3 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.
- 9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	$\text{CGPA} \geq 7.5 \text{ \& } \leq 10.0$
First Class	$\text{CGPA} \geq 6.0 \text{ \& } < 7.5$
Second Class	$\text{CGPA} \geq 5.0 \text{ \& } < 6.0$
Pass Class	$\text{CGPA} < 5.0$

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF TECHNOLOGY & ENGINEERING

A. Choice Based Credit System:

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A 'Core Course' is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An 'Elective Course' is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE)

A 'Programme Elective Course' is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time.

1.2.3 Cluster Elective Course (CE)

A 'Institutional Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non Credit Courses.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
TEACHING AND EXAMINATION SCHEME FOR M Tech in Electrical Engineering (Electrical Power Systems)
CHOICE BASED CREDIT SYSTEM SCHEME (ACY: 2019-2020)

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Theory	Practical/ Tutorial	Total	Credit	Theory		Practical		Total
							Internal	External	Internal	External	
Sem 1	EE741	Modeling & Simulation of Electrical Machines (PC - 1)	4	2	6	5	30	70	25	25	150
	EE742	Modern Power System Operations (PC - 2)	3	2	5	4	30	70	25	25	150
	EE743	Advanced Power Electronics in Power Systems (PC - 3)	3	2	5	4	30	70	25	25	150
	EE7XX	Programme Elective-1 (PE - 1)	3	2	5	4	30	70	25	25	150
	EE7XX	Programme Cluster Elective - 1 (CT - 1)	3	2	5	4	30	70	25	25	150
	XXXX	University Elective 1 (UE - 1)	0	2	2	2			30	70	100
	HSXXX	HSS Elective -I	2		2	2	30	70	0	0	100
			18	12	30	25	1800	420	155	195	950

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Theory	Practical/Tutorial	Total	Credit	Theory		Practical		Total
Sem 2	EE744	Advanced Power System Dynamics & Stability (PC - 4)	4	2	6	5	30	70	25	25	150
	EE745	FACTS & HVDC Transmission (PC - 5)	3	2	5	4	30	70	25	25	150
	EE746	Advanced Power System Protection (PC - 6)	3	2	5	4	30	70	25	25	150
	EE7XX	Programme Elective - 2 (PE - 2)	3	2	5	4	30	70	25	25	150
	EE7XX	Programme Cluster Elective - 2 (CT - 2)	3	2	5	4	30	70	25	25	150
	HS705A	Academic Writing	0	2	2	2	0	0	30	70	100
	XXXX	University Elective 2 (UE - 2)	0	2	2	2	0	0	30	70	100
			16	14	30	25	150	350	185	265	950

Programme Electives of Semester 1 and 2

Programme Electives 1 (PE 1) for Semester 1			Programme Electives 2 (PE 2) for Semester 2		
EE761	Control System Engineering		EE763	Modeling and Simulation of Renewable Energy Sources	
EE762	Engineering Optimization Techniques		EE764	Distributed Generation and Smart Grid	

Cluster Electives of Semester 1 and 2

Cluster Electives 1 (CT 1) for Semester 1			Cluster Electives 2 (CT 2) for Semester 2		
EE771	Digital Signal Processing & its Application		EE772	Restructuring & Deregulation of Power Systems	
EC767	Embedded System Design		EC770	Digital Image & Video Processing	
CE767	Operating System Design & Concepts		CE768	Software Project Management & Quality Assurance	

University Electives of Semester 1 and 2			
University Electives (UE 1) for Semester 1		University Electives (UE 2) for Semester 2	
MA771.01	Reliability and Risk Analysis	EE782.01	Energy Audit and Management
EE781.01	Optimization Techniques	CE771.01	Project Management
ME781.01	Occupational Health and Safety	CA 842	Mobile Application Development
CE772.01	Research Methodology	PT796.01	Fitness & Nutrition
CA730	Internet & Web Designing	NR752.01	Epidemiology and Community Health
PT795.01	Health & Physical Activity	OC733.01	Introduction to Polymer Science
NR 755	First Aid and Life Support	MB651	Software based Statistical Analysis
RD701.01	Introduction to Analytical Techniques	PH892	Intellectual Property Rights
RD702.01	Introduction to Nanoscience& Technology	MA772.01	Design of Experiments
MB650	Creative Leadership	PD262	Astrophysics, Space and Cosmos-I
PH891	Community Pharmacy Ownership		
PD261	Astrophysics, Space and Cosmos-I		

HSS Elective for Semester 1	
HS703.01 A	Languages (French/ German)
HS704 A	Academic Speaking

Semester	Course Code	Course Title	Credit	Examination Scheme					
				Internal		External			Total
				Progress Report	Progress Seminar	Report	Seminar	Viva	
Sem 3	EE811	Project Preliminaries/Summer Industrial Training	4	50	50	0	50	50	200
	EE812	Project Phase 1	18	100	100	100	100	100	500
			22	150	150	100	150	150	700
Sem 4	EE813	Project Phase 2	30	200	200	200	200	200	1000
			30	200	200	200	200	200	1000
		Grand Total	102						3600

M. Tech. (Electrical Power System) Programme

SYLLABI (Semester – I)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE741: MODELING & SIMULATION OF ELECTRICAL MACHINES (PC-1)**M.TECH IST SEMESTER****A. Credit & Hours:**

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Min. No. of Hrs
1	Basics of Electrical Machines Analysis.	10
2	Reference Frame Theory & Transformation.	09
3	Modeling of DC machines	09
4	Modeling of Induction Machines	11
5	Modeling of Synchronous Machines	11
6	Vector Control for Drives	10

Total hours (Theory): 60

Total hours (Lab): 30

Total hours:90

C. Detailed Syllabus:

- 1. Basic of Electrical Machines Analysis** **10 Hours 16.67 %**
Introduction, Magnetically coupled Circuits, Linear Magnetic System, Non Linear Magnetic System, Electromechanical Energy Conversion, Energy Relationships, Energy in Coupling Fields, Graphical Interpretation of Energy Conversion, Machine Windings and Air Gap MMF, Conventions, The Basic Two Pole Machine, Transformer and Speed Voltages in the Armature, Kron's Primitive Machine.
- 2. Reference Frame Theory & Transformation in Machines** **09 Hours 15.00%**
Background of Reference Frame Theory, Equations of Transformation, Stationary Circuit Variables Transformed to the Arbitrary Reference Frame: Resistive Element, Inductive Element, Capacitive Element, Commonly Used Reference Frame, Transformation between Reference Frames, Transformation of Balanced Set, Power Invariance, Balanced Steady State Phasor Relationships
- 3. Modeling of DC Machines** **09 Hours 15.00%**
Introduction, Voltage and Torque Equations, Basic Types of Direct-Current Machines: Separately Excited, Shunt, Series And Compound Machines, Numerical, Dynamic

Characteristics of PMDC and Shunt DC Motor, Time Domain Block Diagrams and State Equations

4. **Modeling of Induction Machines** 11 Hours 18.33 %
Voltage Equations in Machine variables, Torque Equation in Machine variables, Equations of Transformation for Rotor Circuits, Voltage Equations in Arbitrary Reference Frames Variables, Torque Equation in Arbitrary Reference Frame Variables, Equations of Induction Machine using Synchronous Reference Frame with Currents as Variables with Rotor Flux as Variables, Linearization of Machine Equations, Block Diagram for Computer Simulation
5. **Modeling of Synchronous Machines** 11 Hours 18.33 %
Voltage Equations in Machine Variables, Torque Equation in Machine Variables, Stator Voltage Equations in Arbitrary Reference Frame, Voltage Equations in Rotor Reference Frame, Block Diagram for Computer Simulation, Torque Equations in Substitute Variables, Rotor Angle and Angle between Rotors
6. **Vector Control for Drives** 10 Hours 16.67 %
Vector Control Analogy with Separately Excited DC Motor Drive, Vector Control Principle by Phasor Diagram and its Implementation, Induction Motor Flux Estimation Methods, Voltage Model Feedback Signal Estimation of Induction Motor, Current Model Feedback Signal Estimation, Direct or Feedback Vector Control with Rotor Flux Orientation, Indirect Vector Control Phasor Diagram and Equations, Indirect or Feed Forward Vector Control with Rotor Flux Orientation, Phasor diagram for Stator flux Oriented Vector Control, Stator Flux Oriented Vector Control Block Diagram with Estimation Equations.

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Learn the basic modelling concepts of machines.
CO2	Simulate and control the electrical machines in software.
CO3	Learn the generalized theory of machines, reference frame theories and various types of transformation which will help then to analyse the machines.
CO4	Transient analysis of machine performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	2	-	-	1
CO2	2	3	2	1
CO3	3	2	-	2
CO4	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Books:

1. Paul. C. Krause, Oleg Wasynczuk, Scott . D. Sudhoff, **Analysis of Electric Machinery and Drive Systems**, 2nd Edition, IEEE Press A John Willey Inter science.
2. Dr. P.S.Bimbhra, **Generalized theory of electrical machines**, Khanna Publishers.
3. A.E Fitzgerald, Charles Kingsley, Stephen .D. Umans, **Electric Machinery**, 6th Edition, TMH.
4. Bimal Bose, **Power Electronics and Motor Drives Advances and Trends**, Academic Press, Elsevier Publications.

EE742: MODERN POWER SYSTEM OPERATIONS (PC-2)**1ST SEMESTER AND 1ST YEAR****M.TECH (Electrical Engineering)****A. Credit Hours:**

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Hrs
1	Economic dispatch (ED) of thermal generating units and methods of solution	08
2	Transmission System Effects	08
3	Unit Commitment (UC)	05
4	Automatic Generation Control (AGC)	04
5	Optimal Power Flow (OPF)	04
6	Power System Security	05
7	State Estimation in Power System	08
8	Demand Side Management (DSM)	03

Total hours (Theory) : 45

Total hours (Lab) : 30

Total hours : 75

C. Detailed Syllabus:

1. Economic dispatch (ED) of thermal generating units and methods of solution 08 Hours 18%

Basic load modeling concept, Thermal system dispatching with network loss consideration, Lambda iteration method, Gradient method for ED, Newton's method for ED, ED with piecewise linear cost function, ED using dynamic programming

2. Transmission System Effects 08 Hours 18 %

The power flow problem and its solution, The formulation of AC power flow, Gauss-

Siedel method, Newton-Raphson method, Decoupled power flow, Transmission loss calculation, Two generators system, Coordination equations, incremental losses and penalty factors, B – Matrix loss formula, Exact methods of calculating penalty factors, Discussion of reference bus Vs load center penalty factors, Reference bus penalty factors direct from AC power flow

- | | | |
|--------------------------------|--------------|-------------|
| 3. Unit Commitment (UC) | 05 | 11 % |
| | Hours | |

Constraints in UC, Spinning reserves, Thermal unit constraints, UC solution methods, Priority list methods, Forward DP approach, Lagrange relaxation solution

- | | | |
|--|--------------|------------|
| 4. Automatic Generation Control (AGC) | 04 | 9 % |
| | Hours | |

Active power and frequency control, Generator model, Speed governor model, Load model, Speed regulation or droop characteristic, Generation control, Supplementary control action, Tie-line control, Generation allocation, Impact of excitation system on AGC performance, AGC in restructured power system, Availability based tariff (ABT) for frequency regulation in India

- | | | |
|------------------------------------|--------------|------------|
| 5. Optimal Power Flow (OPF) | 04 | 9 % |
| | Hours | |

Solution methods for OPF, Security constrained optimal power flow

- | | | |
|---------------------------------|--------------|------------|
| 6. Power System Security | 05 | 11% |
| | Hours | |

Operating states of power system, Factors affecting power system security, Contingency analysis, Overview of security analysis, Linear sensitivity factors, AC power flow methods, Contingency relaxation

- | | | |
|--|--------------|-------------|
| 7. State Estimation in Power System | 08 | 18 % |
| | Hours | |

Power system state estimation: Past, Present and Future, Weighted least square estimation

Maximum likely-hood concept, Matrix formulation, State estimation of AC network, Detection and identification of bad data measurement, Estimation of quantities not being measured, Network observability and Pseudo measurement, Application of state estimation, Addition of PMU measurements to state estimator, PMU placement for incomplete observability

8. Demand Side Management (DSM)**03 Hours****7 %**

Benefits of DSM, Concepts and Methods of DSM, Load Control, Energy Efficiency, Load Management, DSM Planning, Design, Marketing, Impact Assessment, Cost benefit analysis and feasibility of DSM program, DSM Program Costing and Load Shape Impact on System, DSM Program Cost/Benefit and Feasibility, Environmental Benefits

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of modern power systems and formulations of load flow problems.
CO2	To create and apply appropriate solution methods to solve power system problems
CO3	Apply latest Artificial intelligence methods to solve OPF problems

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	-	-	1
CO2	-	-	2	2
CO3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.

- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- At the end of course, the students will acquire the knowledge regarding the optimization of power system and tools to operate the power system with minimum cost along with ensuring the security.

F. Recommended Study Material:

Text Books

1. A. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, John Wiley & Sons, New York, USA, 1996.
2. Grainger and Stevenson, Power System Analysis, McGraw Hill
3. J.S. Dhillon and D.P. Kothari, Power System Optimization, PHI, 2004
4. Hadi Saadat, Power System Analysis, McGraw Hill
5. Chakrabarti & Halder, Power System Analysis Operation And Control, PHI
6. Ramana N.V., Power System Operation & Control, Pearson Education India
7. M. E. El-Hawary, Gustav S. Christensen, Optimal economic operation of electric power systems, Academic Press
8. Jizhong Zhu, Optimization of Power System Operations, John Wiley & Son

EE743: ADVANCED POWER ELECTRONICS IN POWER SYSTEM
1st Semester and 1st Year
M. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Power Electronics	7
2	Modulation Techniques	7
3	Operation of Various Converters	10
4	Active Filters	10
5	Fundamentals of Harmonic Analysis	7
6	Power Quality Problems & Standards	4

Total hours (Theory) : 45 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 75 Hrs

C. Detailed Syllabus:

1	Basics of Power Electronics.	15%	07 Hrs
	Single phase full bridge converter with R,R-L-E load, analysis of two pulse bridge converter with continuous conduction, Three phase full bridge converters with R,R-L-E load, Buck and Boost DC-DC Converter, voltage source inverter(180°mode) and current source inverter, Turn ON Snubber and Turn off snubber circuits.		
2	Modulation Techniques	15%	07 Hrs
	Single PWM Technique, Multiple PWM technique, Sinusoidal PWM technique (Unipolar and Bipolar), selective Harmonic Elimination, Voltage Space Vector Modulation Technique		
3	Operation of Converters	22%	10 Hrs
	Multi pulse converter: Y/Z phase shifting transformers, Δ/Z phase shifting transformer, Definition of THD and PF, 12 pulse and 18 pulse series type diode rectifier		

	Multi-Level Inverter: multilevel concept, diode clamped multilevel inverter: Principle of operation and features, Flying capacitor multilevel inverter: Principle of operation and features, Cascaded multilevel inverter: Principle of operation and features Single phase Matrix Converter: Principle of operation		
4	Active Filters.	22%	10 Hrs
	Concept and evolution of electrical Power Theory, Sources of Harmonic Production, Power definitions in Sinusoidal & Non Sinusoidal case Instantaneous reactive Power theory, Harmonic elimination by use of Active Filters, Power Circuit Diagram & Control Block diagram for Active Series, Shunt & Hybrid Filters		
5	Fundamental of Harmonic Analysis.	15%	07 Hrs
	Fourier analysis, Representation of Harmonics using Fourier Series expansion with Numerical/Example, Sequence of Fundamental, Fourth & Seventh Harmonic, Sequence of Second, Fifth & Eighth Harmonic, Absence of Triplen Harmonics in Line voltage. Harmonic Phase sequence in balanced Three Phase Power System Measurement of Harmonic Distortion: RMS Voltage & Current, Voltage & Current Distortion Factors, Relation of Power in Resistance, Inductance & Capacitance with Total Harmonic Distortion, Definition of Fundamental & Distortion Power Factor, Power Factor & THD, Current & Voltage Crest Factor. Source & Effects of Harmonic in Power System: sources of Harmonics, Transformers, Rotating machines, Fluorescent lights, Cycloconverters. Effect of Harmonic Distortion on Power system: Thermal, Iron & Dielectric Losses, Harmonics effect on Power system Harmonic Filter Design: Hardware configuration of various Filters, Series Tuned Filter Second order Damped Filter, Band Pass Filters to be connected at AC side of HVDC Converters. Limits of Harmonic Distortion		
6	Power Quality problems & Standards	11%	04 Hrs
	Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. Brief Introduction of Power quality Standards: IEC 61000-2-5, IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)		

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Enhance their basic skills in applications of Power Electronic converters in Power systems.
CO2	Know the performance and classification of Power Electronic converters.
CO3	Understand the Harmonics and its effect on power system applications.
CO4	Design of Active filters and hardware of Power electronics converters.
CO5	Hardware implementation of Power Electronic Converters and its control strategies.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	2	-	1	-
CO2	2	-	-	1
CO3	2	-	-	1
CO4	3	-	2	2
CO5	3	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Dr.P.S.Bimbhra, Power Electronics, Khanna publishers
2. M.H. Rashid, Power electronics handbook, Academic press
3. Power Electronics Essentials & Applications L Umanand, Wiley Publications
4. Bin Wou, High power converters and AC drives, IEEE press
5. Akagi, Watanabe, Aredes, Instantaneous power theory and applications to power conditioning, IEEE press
6. George J. Wakileh, Power System Harmonics, Fundamentals, Analysis and Filter Design, Springer International.
7. J. Arrillaga, N.R. Watson, Power System Quality Assessment, John Willey & Sons Publication

Web Material:

- <https://nptel.ac.in/courses/108/107/108107157/>
(Videos and notes by Prof. Avik Bhattacharya)
- <http://www.nptel.ac.in/courses/108105066/>
(Notes from IIT Kharagpur)
- <http://www.nptel.ac.in/courses/108101038/1>
(Videos and Notes by Prof. B.G. Fernandes, IIT Bombay)
- <http://www.nptel.ac.in/courses/108108035/1>
- <http://www.nptel.ac.in/courses/108108036/>
(Videos and Notes by Prof. L. Umanand, IIT Bangalore)
- <http://www.nptel.ac.in/courses/108102046/>
- <http://www.nptel.ac.in/courses/108108077/>
(Videos and Notes by Prof. K. Gopakumar, IISC, Bangalore)

EE761: CONTROL SYSTEM ENGINEERING(PE-1)**M.TECH IST SEMESTER****A. Credit & Hours:**

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Min. No. of Hrs
1	Time and Frequency Response Analysis	10
2	State Space Analysis	12
3	PID Controllers and Modified PID Controllers	06
4	Digital Control System	10
5	Advances in Control System	07

Total hours (Theory): 45

Total hours (Lab): 30

Total hours:75

C. Detailed Syllabus:**1. Time and Frequency Response Analysis: 10 Hours 22%**

Introduction to first, second order system, Determination of stability using R-H Criterion, Root Locus Method and Bode Plot. Error coefficients, Experimental Determination of Transfer Function, Lead Compensation, Lag Compensation, Lag-Lead Compensation and Parallel Compensation.

2. State Space Analysis 12 Hours 27%

Introduction to State space concept, State space representation of Transfer Function, Concept of Controllability and Observability, Kalman's test for Controllability and Observability, Similarity transformation matrix from Controllability and Observability matrix, Concept of Transfer matrix. Pole placement design through State Feedback, State Observer.

3. **PID Controllers and Modified PID Controllers:** **06 Hours 13%**
 Introduction, Ziegler-Nichols Rules for Tuning PID Controllers, Design of PID Controllers with Frequency-Response Approach, Design of PID Controllers with Computational Optimization Approach, Modifications of PID Control Schemes, Two-Degrees-of-Freedom Control, Zero-Placement Approach to Improve Response Characteristics.
4. **Digital Control System:** **10 Hours 22%**
 Discrete-time systems, Sampled-data and digital control system, Digital controller v/s analog controller, Sampling process, Sample and hold device, Frequency response of Zero-order hold, z-transform, Pulse transfer function of cascaded elements, Pulse transfer function of closed loop systems, Jury's stability test.
5. **Advances in Control System :** **07 Hours 16%**
 Robust control system, Optimal control system, Nonlinear control system, Liapunov Stability analysis of Nonlinear control system, Concept of h_∞ infinity.

D. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

E. Course Outcomes (COs)

Upon successful completion of this course, a student will be able to

CO1	Determine the time and frequency-domain responses of first and second-order systems to step and sinusoidal (and to some extent, ramp) inputs with controller
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	designs.
CO2	Express and solve system equations in state-variable form (state variable models).
CO3	To acquire the skills related to stabilize the systems with different types of controllers and its applications for the linear and nonlinear control systems.
CO4	To solve the physical and mathematical models of the systems with the help of digital tools of control system in a systematic way.
CO5	Learn advancement in control system for linear and nonlinear control system with the help of advanced mathematical tools.

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	2	-	1	3
CO2	3	-	1	2
CO3	3	-	2	2
CO4	3	-	2	2
CO5	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Books

1. Katsuhiko Ogata, Modern Control Engineering, Prentice-Hall
2. D. Roy Choudhury, Modern Control Engineering, PHI
3. S K Bhattacharya, Control systems engineering , Pearson Education
4. Benjamin Kuo, Digital Control System, Oxford University Press
5. M Gopal, Digital control and state variable methods, TMH
6. A. Anand Kumar, Control Systems, PHI

EE762: ENGINEERING OPTIMIZATION TECHNIQUES (PE-1)**1ST SEMESTER AND 1ST YEAR****M.TECH (Electrical Engineering)**

A. Credits Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr No.	Title of Unit	Min. no of hours
1	Fundamentals of Optimization	02
2	Linear Programming	07
3	Unconstrained optimization	08
4	Nonlinear Programming	09
5	Fundamentals of Artificial Intelligence methods	02
6	Applications of Particle Swarm Optimization and Cuckoo Search Algorithm for Power System Problem	10
7	Artificial Neural Network and Fuzzy logic.	05
8	Applications of Hybrid AI methods	02

Total Hours (Theory) :45

Total Hours (Lab) : 30

Total Hours: 75

D. Detailed Syllabus

- | | | | |
|----------|--|-----------------|-------------|
| 1 | Fundamentals of Optimization | 02 Hours | 4 % |
| | Introduction, Feasibility and optimality, Convexity, constraints, Rates of convergence | | |
| 2 | Linear programming | 07 Hours | 16 % |
| | Introduction, Formulation of Linear programming problem, Graphical method, Simplex method, Basic solution, Basic feasible solution, Simplex algorithm, Big M method (Penalty method), Two phase method, Special cases in simplex method application | | |
| 3 | Unconstrained Optimization | 08 Hours | 18 % |
| | Introduction, Optimality conditions, Newton's method for minimization, Line search methods, Steepest-Descent method, Quasi-Newton method, Golden section method | | |
| 4 | Nonlinear Programming | 09 Hours | 20% |
| | Optimality conditions for constrained problems, Kuhn-Tucker conditions, Penalty function method, Barrier method, The Lagrange multipliers and the Lagrangian function, Sensitivity analysis, Computing the Lagrange multipliers, Sequential quadratic programming, Interior point method | | |
| 5 | Fundamentals of Artificial Intelligence methods | 02 Hours | 4 % |
| | To understand importance of AI methods and their comparison with various classical methods using various criteria. | | |
| 6 | Applications of Particle Swarm Optimization and Cuckoo Search Algorithm for Power System Problem | 10 Hours | 22 % |
| | Introduction to PSO, Unconstrained and constrained optimization using PSO, Effects of various coefficients on convergence, Cuckoo Search (CS) Algorithm, PSO and CS for Economic dispatch problem, comparison between PSO and CS | | |
| 7 | Artificial Neural Network and Fuzzy logic. | 05 Hours | 11 % |
| | Introduction to ANN, Supervised learning, Factors affecting the performance of ANN models, Back-Propagation Learning, Introduction to Fuzzy logic and characteristics of Fuzzy sets, Fuzzy relation, De-fuzzification methods. | | |
| 8 | Applications of Hybrid AI methods | 02 Hours | 5 % |
| | Advantage of hybrid AI methods, Application of Hybrid GA-Non-linear Programming in optimal relay coordination, Hybrid Interior point method, PSO for obtaining LMP. | | |

E. Course Outcomes (COS):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of optimization and formulations of optimization
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	problems.
CO2	To create and apply appropriate solution methods to solve various optimization problems.
CO3	Use research base knowledge and apply Artificial intelligence methods to solve OPF problems
CO4	Use of latest IEEE winner optimization methods to solve complex engineering problems.

Mapping of course outcomes with program outcomes
Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	2	3	-	1
CO2	-	-	3	2
CO3	-	2	-	-
CO4	1	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- At the end of the course, the students will acquire the knowledge regarding various classical and Artificial intelligence optimization methods. They will be able to solve

various real world power system optimization problems using various optimization methods. Such skill will be beneficial in getting placement in various power sectors.

G. Recommended Study Material:

Books:

1. M. E. El-Hawary, Electric power applications of Fuzzy systems , IEEE press, 2006
2. N.P.Padhy, Artificial intelligence and intelligent systems , Oxford University Press, 2005
3. Chaturvedi, Soft Computing Techniques and its Applications in Electrical Engineering , Springer
4. Narayan Rao, Optimization Principles, John Wiley & Sons
5. K. Deb, Optimization for engineering design, PHI

EE771: DIGITAL SIGNAL PROCESSING AND IT'S APPLICATIONS (CT-1)

M. TECH Ist SEMESTER

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the course:

Sr. No	Title of Unit	Hrs
1.	Introduction to Signals and Systems	10
2.	Realization of Discrete time systems	10
3.	Discrete Fourier Transform	10
4.	Digital Filter Design	15

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

C. Detailed Syllabus:

- 1. Introduction to Signals and Systems** 10 Hours 22 %
Basic elements of Digital Signal Processing System. Advantages of Digital over Analog Signal Processing, Generalized Block diagram of a DSP System, Continuous-time Vs Discrete time signals, Classification of discrete time Signals, Deterministic and Random Signals, Periodic and A periodic signals, Even and odd signals, Exponential and Sinusoidal signals, Causal and Non-Causal signals, Energy and Power Signals, Some Standard signals like Impulse, Step, Ramp and Parabolic. Basic operations on signals: Shifting, Scaling, Time reversal or Folding, Addition, Subtraction, Multiplication. Classification of discrete time Systems: Static and Dynamic Systems, shift-variant and invariant systems, Linear and Non Linear systems, Causal and Non Causal systems, Stable and Unstable systems, impulse response and convolution sum, analytical evaluation of discrete convolution, properties of convolution
- 2. Realization of Discrete Time Systems** 10 Hours 22 %
Introduction to Z Transform, properties of Z transform, Numerical for Z transform,

Realization of discrete time systems: basic building block, types of digital system, **Structures for Realization of IIR Systems and numerical:** Direct Form I, Direct Form II, Cascade form, Parallel form, **Structures for Realization of FIR systems and numerical:** Direct Form, Cascade Form Realization

3. **Discrete Fourier Transform** 10 Hours 22%

Concepts of frequency in continuous-time and discrete time signals, Discrete Fourier Series: Exponential form, trigonometric form, relation between exponential and trigonometric form of discrete Fourier series, properties of discrete Fourier transform (DFT), Examples, Relation between DFT and Z transform. Numerical to compute DFT of signal, Computational Advantage of Fast Fourier Transform(FFT) Over DFT, Decimation in time (DIT) & Decimation in Frequency (DIF), Algorithm for implementation of FFT: Derivation, Butterfly Diagram & Related Numerical.

4. **Digital Filter Design** 15 Hours 34 %

Advantages and disadvantages of digital filter over analog filters,
Classification of filter: low pass, high pass, band pass and band reject (Band Stop), filter specification and Magnitude Characteristics of Low Pass Filter.
Design of IIR filters from analog filters: derivation for impulse invariant & Bilinear transformation method, Analog Butterworth and Chebyshev Filters, Design of digital low pass Butterworth filter and low pass Chebeshev filters, Design of High Pass, Band Pass and Band Reject IIR Filters from Low Pass IIR Filter, Frequency transformation in Analog Domain and Frequency transformation in Digital Domain & Related Numerical
Design of FIR Filters: Advantages and Disadvantages of FIR Filter, design of FIR filters using windows, various windows for FIR filter and summary, frequency sampling techniques, numerical

D. Course Outcomes (COs):

At the end of this course, the student should:

Upon successful completion of this course, a student will be able to

CO1	Have good understanding about basic processing and realization of Digital Signal and system. Hence, The students will be able to understand basic terms and keywords related to signal processing in research paper and will be able to describe their
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	algorithm in the required mathematical form/nomenclatures.
CO2	To understand Transforms and application for Digital Signal Processing. The Fourier transform and Z transform are targeted here and their application to Power system will be understood.
CO3	To understand theory and application of Infinite Impulse Response (IIR) filters and be able to design IIR filters.
CO4	To understand theory and application of Finite Impulse Response (FIR) filters and be able to design IIR filters.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	3	-	3
CO2	2	-	-	3
CO3	3	2	-	3
CO4	3	2	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a component of the overall evaluation.
- Minimum two internal exams will be conducted and will be considered as a part of overall evaluation.
- Assignments based on course content will be given to the students for each unit/topic and will be evaluated at regular interval and its weightage may be reflected in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Books:

1. V. Udayashankara: Real time Digital Signal Processing, Fundamentals, Algorithms and Implementation using TMS Processor, Eastern Economy Edition
2. A. Anandkumar, Digital Signal Processing, PHI Learning Pvt Ltd.

References:

1. John G. Proakis, Dimitris Manolakis: Digital Signal Processing - Principles, Algorithms and Applications, Pearson, ISBN 0-13-394289-9
2. Steven T. Karris, Signals and Systems with MATLAB Computing and Simulink Modelin, Orchard Publications.

Web Material:

1. <http://nptel.iitm.ac.in/video.php?subjectid=117102060> nptel video course by prof.s.c. duttaroy, iitdelhi.
2. DSP Video Lectures by Prof. E. Ambikairajah:
http://onlinevideolecture.com/?course_id=373
3. http://www.analog.com/en/processorsdsp/processors/beginners_guide_to_dsp/fca.html a beginner's guide to digital signal processing

EC767: Embedded System Design (CT-1)

M. Tech Ist SEMESTER

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Embedded Systems	08
2	Typical Embedded Systems	14
3	RISC microcontrollers	18
4	Embedded product development life cycle	05

Total Hours (Theory):45

Total Hours (Lab): 30

Total Hours:75

C. Detailed Syllabus:

This will provide details about topics under each units of the course.

1	Introduction to Embedded Systems	8Hrs	18%
1.1	History of embedded systems		1hr
1.2	Classification of embedded systems		1hrs
1.3	Major application area of embedded systems		2hrs
1.4	Purpose of embedded systems		1hrs
1.5	Fundamental issues in hardware software co-design		1 hrs
1.6	Introduction to unified modeling language (UML)		2 hrs
2	Typical Embedded Systems	14Hrs	30%
2.1	Core of the Embedded Systems		2hrs
2.2	Memory		3hrs

2.3	Sensors and actuators		4hrs
2.4	Communication interface		3hrs
2.5	Embedded firmware		2hrs
3	RISC microcontrollers	18Hrs	40%
3.1	RISC and CISC architectures		1hrs
3.2	AVR architecture and pin functions		2hrs
3.3	AVR programming in C		3hrs
3.4	I/O interfacing: LED, multiplexed 7-segment, LCD, GLCD, sensors, keypad, relay, buzzer		5hrs
3.5	AVR interrupt programming in C		2hrs
3.6	AVR serial programming in C		2hrs
3.7	Communication protocol: I2C protocol and RTC interfacing, SPI protocol and max7221 interfacing		3hrs
4	Embedded product development life cycle	5Hrs	12%
4.1	Product enclosure design tool		1 hrs
4.2	Product enclosure development techniques		1 hrs
4.3	Objective of EDLC		1 hrs
4.4	Different phases of EDLC and approaches		2 hrs

D. Instructional Method and Pedagogy:

- Multimedia Projector
- OHP
- Chapter wise Assignments
- Quiz
- Chalk + Board
- White Board
- Online Demo

E. Recommended Study Material:

❖ Text Books:

1. Introduction to Embedded Systems by shibu K V mcgraw hill
2. The AVR microcontroller and Embedded Systems by muhammad Ali Mazidi, Sarmad Naimi, Sepehr Naimi

❖ **Reference Books:**

1. System Design: A Unified Hardware/Software Introduction by Frank Vahid and Tony D. Givargis, Addison Wesley, 2002.
2. Computers as Components by Wayne Wolf, Morgan Kaufmann, 2001
3. Embedded C programming and the ATMEL AVR by Barnett, cox and o'cull, Thomson

CE767: OPERATING SYSTEM DESIGN & CONCEPTS (CT-1)
M.TECH IST SEMESTER

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Basics of operating system	02
2.	The Process and The kernel	08
3.	Threads and Lightweight Processes	04
4.	The Buffer Cache	06
5.	System Calls for the file system	08
6.	Signal and Session Management	06
7.	Interposes Communications	06
8.	Case Study: Multiprocessor Systems, Distributed Unix Systems	05

Total hours (Theory): 45 Hrs.

Total hours (Lab): 30 Hrs.

Total hours: 75 Hrs.

C. Detailed Syllabus:

- Basics of operating system** 02Hours 4%
Introduction to an operating system, history of an Operating system, computer hardware review, Operating system Concepts, System Calls
- The Process and The kernel** 08 Hours 18%
Introduction, architecture of Unix operating system, Mode space and context, The process abstraction, kernel data structure, Executing in kernel mode, System administrator
- Threads and Lightweight Processes** 04Hours 8%

Process control, Fundamentals abstraction, Lightweight Process design, User-level threads libraries, Multithreading in Solaris and SVR4, Threads in Mach, Digital Unix, Mac 3.0 continuations

4. The Buffer Cache 06Hours 14%

Buffer headers, Structure of the buffer pool, Scenario for retrieval of a buffer, Reading and writing disk blocks, Advantages and disadvantages of duffer cache.

5. System Calls for the file system 08Hours 18%

Open, read , write , file and record locking, adjusting the position of File I/O - lseek , close, file creation, creation of special file, change directory and change root, change owner and change mode, STATE and FSTATE, Pipes, Dup, mounting and unmounting file systems, link and unlink, file system abstractions and maintenance

6. Signal and Session Management 06Hours 14%

Single generation and handling, Unreliable single, Reliable single, Singles in SVR4, Signals implementations, Exceptions, Mach exception handling, Process groups and Terminal Management, The SVR4 sessions architecture

7. Interprocess Communications 06Hours 14%

Universal IPC facilities, System V IPC, Mach IPC, Messages, Ports, Message passing, Port operations, Extensibility Mach 3.0 enhancements, discussion

8. Case Study: Multiprocessor Systems, Distributed Unix Systems 05Hours 10%

overview, Solutions with master and slave processors, solutions with semaphore, performance limitation, Satellite Processors, The Newcastle connection, transparent distributed file systems

D. Instructional Method and Pedagogy:

- A laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Minimum 10 experiments are suggested in the laboratory related to course content.

E. Student Learning Outcome:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Faculty would use coached problem solving method as it is class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.

- Attendance is compulsory in lectures and laboratory which carries 5% component of the overall evaluation.
- Internal exams or Open-book tests will be conducted and average will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weight age of 5%.
- Surprise tests/Quizzes will be conducted which carries 5% component of the overall evaluation.

The course includes

- Students will be able to learn advanced topics in design and implementation of microkernel-based.
- Students will also learn to read and critique research papers.
- Students will be familiar with classic operating systems literature.
- Students will make substantial contributions to the operating systems project.

F. Recommended Study Material:

❖ Text Books:

1. "UNIX Internals" by Uresh Vahalia Prentice Hall Press
2. "The Design of the Unix Operating System" by Maurice J. Bach Tata McGraw Hill

❖ Reference Books:

1. Advanced Concepts in Operating Systems, Singhal and Niranjana G.Shivaratna.
2. OS: Advanced Concepts, Maekawa, Oldehoeft. Addison-Wesley.
3. "Distributed Systems", Sape Mullender, Addison-Wesley.
4. Multithreaded Programming with Pthreads, Bil Lewis, Daniel J. Berg.

PG PROGRAMMES

Semester I
HSS Elective

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS703.01 (A/B/C/D/E/F/G/H): LANGUAGES (French)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS703.01	LANGUAGES (French)	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Introduction to French Language Facts and figures about French Language Basic French Linguistics * Alphabets * Accents * Liaison * Nasalization French Culture, Differ between French and English Grammar : • Subject Pronoun • Verbs : (être ,avoir, habiter,regarder,manger ... “er” verb) • Form of address • Numbers (1 to 20) • Nouns and plurals of nouns • The expression : C’est ,Il y a Presentation: 1) Self Introduction 2) Question and answering Dialogue	08
2	Grammar : • Definite articles • Indefinite articles • Present tense ○ Positive Forms ○ Negative Forms • Numbers (21 to 100 , 100-1000) • Days ,Months ,Familly • Verbs: (aller,venir,finir,pouvoir,vouloir “ir “ verb) Social Links: 1) My family & relations 2) Appointments	08

	3) Gathering information from someone Dialogue	
3	Grammar : • Common Adjectives • Comparative Adjectives • Common Adverbs • Interrogative Forms • The expression : “On” • Directions , Countries, Nationalities, Seasons, Weather, • Professions • Verbs: (Prendre, Apprendre, Comprendre, faire ... “re “ verb) Work , Study and Travel 1) Job/ Profession 2) Ticket Reservation (At Bus/At Railway/At Airport) Dialogue	08
4	Grammar : 1) Common Prepositions 2)Common Conjunctions 3)Past Tense 4) Future Tense 5) Colors ,Shapes, Animals ,Vegetables, Fruits 6) Verbs: (“er”, “ir”, ”re” etc...) Food & Shopping 1) Buy a vegetables and fruits 2) Any Conversation between Customer and Vendor (At Mall/ At Restaurant / At Market) Dialogue	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on pragmatic learning. Teaching will be facilitated by Reading Material, Discussion, Task-based learning, assignments and various interpersonal activities like group/pair work, independent and collaborative work, presentations, etc.

IV. Evaluation:

The students will be evaluated continuously in the form of their consistent performance. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

The students' performance in the course will be evaluated on a continuous basis through the following components:

Internal Evaluation:

Sl. No.	Component	Number	Marks per incident	Total Marks
1	Assignment / Presentation/ Task	5	5	25
2	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the LSRW skills in the French Language.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Term Work, Viva and Practical (LSRW)	-	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would:

CO1	Have a basic overview of the French language and culture.
CO2	Develop a basic understanding of the usage of French language such as alphabets,numbers,months , colour etc.
CO3	Learn the basics of French Grammar and their usages
CO4	Learn to use basic language functions such as introducing oneself, answering basic Wh questions
CO5	be able to initiate conversations/dialogues in the French language confidently.

Course Articulation Matrix:

	PO1	PO2	PO3
CO1	-	-	2
CO2	-	-	2
CO3	-	-	2
CO4	-	-	2
CO5	-	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Note: The reference/ reading material will be provided in consultation with the expert.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS704 A | B | C | E | F | G | H: ACADEMIC SPEAKING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS704	Academic Speaking	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Foundations of Advance Communication - Meaning and Definition of Advance Communication - Advance Communication in Digital, Social, Mobile World - Strategies for Advance Communication - Meaning and Concept of Academic Language - High Frequency Academic Vocabulary	04
2	Art of Conversation - Describing people, places and things - Expressing opinions - Making suggesting - Persuading someone - Interpreting and Summarizing	06
3	Science of Power Speaking - Phonemes - Word Stress - Pronunciation - Intonation - Pause - Register - Fluency - Prosody - Lexical Range	06
4	Academic Speaking Application – Part I	08

	• <i>Art of Oratory</i> • <i>Formal Presentation</i> • <i>Speech Analysis – Decoding Best Speeches</i>	
5	Academic Speaking Application – Part II • <i>Job Interview</i> • <i>Group Discussion</i> • <i>Meeting</i>	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

V. Course Outcomes (COs)

After the completion of the course, the student would:

CO1	understand and demonstrate advance communication skills and academic speaking.
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CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3
CO1	1	-	-
CO2	-	2	-
CO3	-	2	-
CO4	-	2	-
CO5	-	2	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

VII. Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

M. Tech. (Electrical Power System) Programme

SYLLABI (Semester – II)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE744: ADVANCED POWER SYSTEM DYNAMICS AND STABILITY
M.Tech. 2nd SEMESTER (Electrical Power System)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	04
2	Synchronous Generator Modelling for Power System Stability	14
3	Excitation System	06
4	Small Signal Stability of Power System	12
5	Transient Stability of Power System	08
6	Voltage Stability of Power System	08
7	Sub-synchronous Oscillations in Power System	06
8	FACTS and HVDC for stability improvement	02

Total hours (Theory) : 60 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 90 Hrs

C. Detailed Syllabus:

1	Introduction	08.00%	04 Hrs
	Introduction to power system fundamentals, Power system control, Power system operating states.		
2	Synchronous Generator Modelling for Power System Stability	23.34%	14 Hrs
	Introduction, Synchronous generator, Generator model, Steady state characteristics- for cylindrical rotor machine: Power factor control, Power angle characteristics, Salient pole synchronous generators. Physical description of synchronous machine, Mathematical description and basic equation of synchronous machine, stator and rotor self and mutual inductances, stator		

	and rotor flux linkages in d-q transformation, stator voltage equations, electrical power and torque, physical interpretation of dq0 transformation, steady state analysis, synchronous machine operation parameters and frequency response characteristics, swing equation		
3	Excitation System	10.00%	06 Hrs
	Need of excitation system, Elements of excitation system, Types of excitation system, DC excitation system, AC excitation system, Static excitation system, Dynamic performance measures of excitation system, Control and protective functions of excitation system, Detail modelling of different excitation systems.		
4	Small Signal Stability of Power System	20.00%	12 Hrs
	Derivation of swing equation, M and H constants, equivalent H constant and examples, Classification of power system stability, Steady state stability, derive the expression for natural frequency of oscillations and damped frequency of oscillations, example, Derivation of synchronizing power coefficient, examples. Eigen value and participation factor analysis, small signal stability of a single machine Infinite bus (SMIB) system, Effect of excitation system and automatic voltage regulator (AVR) on small signal stability, Power System Stabilizer for improvement of damping, Small signal stability of multi-machine systems, Characteristic of small signal stability problems.		
5	Transient Stability of Power System	13.33%	08 Hrs
	Transient stability and factors affecting it, Equal area criteria, Cases of sudden application of mechanical power input, application of three phase fault at the middle of the line or at the end of the line, Example based on equal area criteria, Point by point method for the solution of swing equation and examples, Transient stability of multi machine system, Methods of stability improvement.		
6	Voltage Stability of Power System	13.33%	08 Hrs
	Voltage stability, Classification of Voltage Stability, Reasons for Voltage instability may arise, Transmission System Characteristic, P-V Curve, Q-V Curve, Generator Characteristic, Load Characteristic, Characteristic of reactive power compensating device, Voltage collapse, Prevention of voltage collapse.		
7	Sub-synchronous Oscillations in Power System	10.00%	06 Hrs
	Turbine-generator torsional characteristic, torsional natural frequencies and mode shapes, torsional interaction with power system control like generator excitation, speed		

	governors, nearby HVDC converters, Sub-synchronous resonance (SSR), counter-measures for SSR		
8	FACTS and HVDC for stability improvement	02.00%	02 Hrs

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to acquire the knowledge regarding

CO1	The knowledge regarding to planning and designing of the new interconnected power system with practical simulation.
CO2	Carry out synchronous machine modelling for the power system stability analysis.
CO3	Understand the concept of modeling of the different excitation system.
CO4	Understand different types of stability phenomena being observed in the power system networks including the analysis methods and the design of control measures required for the improvement of system stability with practical simulation.
CO5	Develop the understanding of the Sub-synchronous Oscillations in Power System and enhancement of power system stability using FACTS and HVDC.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSOI
CO1	3	1	1	3
CO2	1	2	-	2
CO3	2	1	-	2
CO4	3	1	1	3
CO5	1	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

F. Recommended Study Material:

Text Book:

1. Power System Analysis by Hadi Saadat, Tata Mcgraw Hill.
2. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
3. Power System Analysis and Design by B.R. Gupta, S. Chand
4. Power System Analysis by Grainger & Stevenson, Tata Mcgraw Hill.

Reference Book:

1. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, John Wiley & Sons, New York, USA.
2. C.W. Taylor, Power System Voltage Stability, McGraw Hill.

Web Material:

1. <http://nptel.ac.in/downloads/108101040/>
2. <http://nptel.ac.in/courses/108104052/2>

EE745 FACTS AND HVDC TRANSMISSION (PC-5)

M.TECH 2nd SEMESTER

A. Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Min. no of hours
1	Characteristic of EHV AC Transmission Network	08
2	FACTS Controllers	14
3	Power Flow with FACTS Controllers	14
4	HVDC Transmission	14
5	Integrated Operation of EHV AC and HVDC Systems	10

Total Hours (Theory) : 60

Total Hours (Lab) : 30

Total Hours: 90

C. Detailed Syllabus

- 1 Characteristic of EHV AC Transmission Network** **08 Hours** **13.33%**
Introduction, uncompensated transmission lines, line equations, SIL, effect of line length, load power and power factor on voltage and reactive power for uncompensated line, maximum power and stability consideration for uncompensated line. Active and passive compensator, uniformly distributed fixed and regulated shunt compensation, lines with mid point shunt and series compensation.
- 2 FACTS Controllers** **14 Hours** **23.33%**
Need of FACTS devices, classification of FACTS devices, operating principle of SVC, V-I characteristic of SVC, advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR, advantages of TCSC,

different mode of operation of TCSC, different modeling concepts of TCSC,
Operating principle of STATCOM, SSSC and UPFC and their applications for power system performance improvement

- | | | | |
|----------|--|-----------------|---------------|
| 3 | Power Flow with FACTS Controllers
Power flow solution with SVC, TCSC, SSSC and UPFC | 14 Hours | 23.33% |
| 4 | HVDC Transmission
HVDC system configuration and components, fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, converter transformer rating, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults | 14 Hours | 23.33% |
| 5 | Integrated Operation of EHV AC and HVDC Systems
Power flow analysis in AC/DC system, modeling of DC links, solution of DC load flow, AC-DC power flow, HVDC light, protection in HVDC, grounding in HVDC systems, stability improvement with HVDC integration | 10 Hours | 16.66% |

D. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to:

CO1	Compare the HVDC Transmission and EHVAC transmission
CO2	Identify, analyze and understand the converter configurations used in HVDC system and its components to enhance the skill development.
CO3	Understand controllers for controlling the power flow through a dc link
CO4	Identify analyze and understand the different FACTS controller required for the power system performance improvement.
CO5	Understand the role of impedance control, phase angle control and voltage control in controlling real and reactive power in transmission systems.
CO6	To learn the mathematical modelling of different FACTS controllers for power flow analysis.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1

CO1	1	1	1	-
CO2	2	2	2	2
CO3	1	-	1	2
CO4	2	2	2	2
CO5	3	1	1	-
CO6	3	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lecturers will be conducted with aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures, which carries 10 marks in overall evaluation.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students for each unit and will be evaluated at regular interval. It carries a weightage of 5 marks in the overall internal evaluation.

F. Student Learning Outcomes:

At the end of the course, the students will acquire the knowledge regarding AC and HVDC transmission network. The students will know the modeling of FACTS and HVDC controllers. They can analyze effectiveness of these controllers for power system improvement.

G. F. Recommended Study Material:

Books and References:

1. E. W. Kimbark, Direct Current Transmission, Vol 1, Wiley-Interscience, 1971
2. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
3. S. Kamakshiah and V. Kamaraju, HVDC Transmission, McGraw Hill, 2011

4. K.R. Padiyar, HVDC Power Transmission Systems: Technology and System Interactions, New Age International
5. E. Acha, FACTS: Modeling and Simulation in Power Networks, Wiley 2004
6. T.J.E Miller, Reactive Power Control in Electric Systems, Wiley India Edition, 1982
7. R. Mohan Mathur and R. K. Varma, Thyristor-based FACTS controllers for Electrical Transmission systems, IEEE Press, 2002

EE746 ADVANCED POWER SYSTEM PROTECTION (PC-6)
2nd Semester M. Tech. (Electrical Power System)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Power System Protection	02
2	Instrument Transformer	04
3	Overcurrent Relay Coordination for Radial and Parallel Feeder	08
4	Basics and Architecture of Numerical Relay	08
5	Phasor Estimation Algorithm	10
6	Synchrophasor and Wide Area Measurement Systems	07
7	Protective Relaying for Abnormal Operating Conditions in Power System	06

Total hours (Theory) : 45 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 75 Hrs

C. Detailed Syllabus:

1	Introduction to Power System Protection	08%	02 Hrs
	Power system structural considerations, Desirable attributes of the protective relaying, Overview of the transmission line and apparatus protection.		
2	Instrument Transformer	12%	04 Hrs
	Current transformer (CT), Saturation characteristic of current transformer, Classification of CT, General rule applicable for using current transformer in electrical networks, examples on CT, CT saturation, caution in CT selection, CT oversizing factor, Voltage transformer (VT), General rule applicable for using voltage transformer in electrical networks, Coupling capacitor voltage transformer (CCVT) and its transient response, Examples.		

3	Overcurrent Relay Coordination for Radial and Parallel Feeder	18%	08 Hrs
	Plug setting multiplier (PSM) and time multiplier setting (TMS), Types of overcurrent relay, Setting of PSM and TMS for relay coordination in radial and parallel feeders, Generalized flowchart for the relay coordination, LINKNET structure		
4	Basics and Architecture of Numerical Relay	18%	08 Hrs
	Need of digital protection, Basic block diagram of numerical relay, ADC, quantization error, ADC types, sampling theorem, need of S/h circuit, aliasing and anti-aliasing filters and its design, examples.		
5	Phasor Estimation Algorithm	20%	10 Hrs
	Mann-Morrison Algorithm, three sample and two sample techniques, differential equation algorithms, solution of differential equation algorithms using numerical integration, application of differential equation algorithm for different line faults, least square Error based algorithm, Generalized Fourier Series (FS) in trigonometry and exponential form, FS for recurring and non-recurring pulse, Discrete Fourier Transform (DFT), samples collection in window, DFT co-efficient, sliding DFT.		
6	Synchrophasor and Wide Area Measurement Systems	12%	07 Hrs
	Introduction, Synchrophasor as per IEEE std C37.118.1-2011, Time Tagging of Synchrophasor, Block diagram of Synchrophasor enabled Digital Relay or PMU, off nominal frequency operation of PMU, Synchrophasor applications.		
7	Protective Relaying for Abnormal Operating Conditions in Power System	12%	08 Hrs
	Digital frequency relays, formulating a load shedding steps and schemes, out-of-step relaying, Effect of out-of-step conditions, Philosophies of out-of-step relaying, Types of out-of-step schemes, Protection of series compensated transmission line, relaying problems due to series compensation, reclosing and synchronizing.		

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of protective relaying principles and to learn how to select the instrument transformers for particular protection scheme.
CO2	Understand and learn how to select the instrument transformers for particular protection scheme.
CO3	Create/Design protection scheme for three phase power transformer, induction motor and bus bar protection with CT and relay connection and settings.

CO4	Develop skill for to understand the different Phasor Estimation Algorithms for numerical relays, Synchrophasor and Wide Area Measurement Systems.
CO5	Recognize the construction and working of numerical relaying. Think about relays in advanced level.

Mapping of course outcomes with program outcomes
Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	1	2	-	1
CO2	1	1	-	1
CO3	3	1	1	2
CO4	3	2	1	2
CO5	1	1	1	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

F. Recommended Study Material:

Text Book:

1. Power System Protection and Switchgear by B.A. Oza, N.C. Nair, R.P. Mehta and V.H. Makwana, Mcgraw Hill, 2010

2. Protection and Switchgear by Bhavesh Bhalja, R. P. Maheshwari and N. G. Chothani, Oxford University Press, 1st Edition, 2011
3. Fundamentals of Power System Protection by Y.G. Paithankar, S.R. Bhide, Prentice Hall, India, 2003

Reference Book:

1. Power System Protection and Switchgear by Badri Ram and D N Vishwakarma, Tata Mcgraw Hill, 2001
2. Power System Protection Static Relays by T.S. Madhavrao, Tata Mcgraw Hill, 1989

Web Material:

- [Online course on Power System Protection by S. A. Soman. \[Online Available\]](http://www.cdeep.iitb.ac.in/nptel/Electrical%20Engineering/Power%20System%20Protection/TOC_M1.html)
http://www.cdeep.iitb.ac.in/nptel/Electrical%20Engineering/Power%20System%20Protection/TOC_M1.html
- [Miroslav Begovic, Damir Novosel, Daniel Karlsson, Charlie Henville and Gary Michel, "Wide-Area Protection and Emergency Control". \[Online Available\]](http://www.science.smith.edu/~jcardell/Readings/TRUST%20US/IEEE%20Proc%20May%202005/WideArea%20Protec%20-%20Begovic.pdf)
<http://www.science.smith.edu/~jcardell/Readings/TRUST%20US/IEEE%20Proc%20May%202005/WideArea%20Protec%20-%20Begovic.pdf>

EE763 MODELING & SIMULATION OF RENEWABLE ENERGY SOURCES (PE-2)
M.TECH 2nd SEMESTER

Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr No.	Title of Unit	Min. no of hours
1	Introduction	03
2	Fixed speed induction generator based wind turbine generating system	08
3	Variable speed turbine generating system	11
4	Modeling of Photovoltaic System and its grid integration	19
5	Dynamic modeling & simulation of fuel cells	04

Total Hours (Theory) : 45

Total Hours (Lab) : 30

Total Hours: 75

B. Detailed Syllabus

- 1 Introduction** **03 Hours** **6.66%**
Indian and global scenario of renewable energy sources in power system, local impacts of renewable energy sources, side wide impact of renewable energy sources on power system, issues of its grid integration, grid code regulations for integration of renewable energy resources.
- 2 Fixed speed induction generator based wind turbine generating system** **08 Hours** **17.77%**
principle of fixed speed wind turbine generating system, FSIG configurations for wind generation: cut in and cut out speed, C_p versus λ characteristics, pitch control, different mass model, induction machine modeling: flux model, current model, voltage

behind transient reactance model, fifth order, third order, first order model, power and torque equations, mathematical model for voltage control of standalone hybrid power system, the MVAR load voltage control problem, excitation control system, the flux linkage equation, incremental reactive power analysis, mathematical model of an induction generator, modeling of a static var compensators, system state space modeling. The integration of wind farms into the power system: reactive power compensation, static var compensator (SVC) and FSIG stability.

- | | | | |
|----------|---|-----------------|---------------|
| 3 | Variable speed wind turbine generating system | 11 Hours | 24.44% |
| | Advantages of DFIG over fixed speed, working principle of DFIG, operation of DFIG in sub synchronous and super synchronous speed, schematic of DFIG and maximum power point tracking (MPPT), the overall control system of a variable speed wind turbine with DFIG, with modeling equations : control scheme of rotor side convertor, control scheme of grid side convertor, System configuration of variable speed FRC synchronous generator based (FRC-SG) Wind Turbine: direct-driven wind turbine generators, permanent magnets versus electrically excited synchronous generators, overall control strategy of PMSG with full scale convertor, control of FRC induction generator based (FRC-IG) Wind turbine. | | |
| 4 | Modeling of Photovoltaic systems & its grid integration | 19 Hours | 42.22% |
| | Photovoltaic Energy scenario in India, Electrical characteristics and working of photovoltaic cell , equivalent circuit for a photovoltaic cell, example of I-V curve for a photovoltaic cell, from cells to a module and from modules to arrays, the PV I-V curve under standard test conditions, impacts of temperature and insolation on I-V curves, shading impacts of I-V curves and Examples, PV I-V curves for different types of loads, Maximum power point tracking (MPPT) algorithms, sun tracking, grid connected systems: interfacing with the utility, grid connected system sizing, battery storage capacity, battery sizing, stand alone pv system design configurations, charge controller for pv system, Hybrid pv system, its design and issues, crystalline silicon technologies: single crystal cz silicon, ribbon silicon technology, cast multicrystalline silicon, crystalline silicon modules, thin film photovoltaics: amorphous silicon, gallium arsenide and indium phosphide, cadmium telluride, copper indium diselenide, Emerging Solar cell technologies: Organic solar cell, Dye-sensitized solar cell. | | |
| 5 | Dynamic modeling & simulation of fuel cells | 04 Hours | 8.88% |

Fuel cell basics, types of fuel cells, comparison of important components and characteristics of different types of fuel cells, fuel cell equivalent circuit, need for fuel cell dynamic models, polymerelectrolyte membrane (PEMFC)dynamic model development: gas diffusion at the electrodes, material conservation, PEMFC output voltage, PEMFC voltage drops, PEMFC model Structure, equivalent electrical circuit model of PEMFC, solid oxide fuel cell (SOFC) dynamic model development: effective partial pressures, material conservation, SOFC output voltage, activation voltage drop, grid-connected system configuration of PEMFC and SOFC.

C. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lecturers will be conducted with aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory, which carries five marks in overall evaluation.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students for each unit and will be evaluated at regular interval. It carries a weightage of 5 marks in the overall internal evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials will be conducted having a share of 5 marks in the overall internal evaluation.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Recognize the electric circuits connected to various Renewable Energy Sources.
CO2	Acquire the knowledge regarding the wind turbines and solar energy system components.
CO3	Understand the construction, working and control of solar and wind energy system.
CO4	Aware with the performance characteristics of Renewable Energy Sources.
CO5	Design & Analyse the performance of PV and wind energy system in simulation and modelling platforms.

Course Articulation Matrix:

	PO1	PO2	PO3	PSO1
CO1	1	-	1	-
CO2	1	-	1	-
CO3	2	-	2	1
CO4	2	1	2	1
CO5	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Books:

1. Wind Energy Generation Modelling and Control, Olimpo Anaya-Lara, Nick Jenkins, Janaka Ekanayake, Phill Cartwright, Mike Hughes, Wiley, A John Wiley & Sons, Inc, Publication.
2. Small Signal Analysis of Isolated Hybrid Power Systems Reactive Power and Frequency Control Analysis, R.C.Bansal, T.S.Bhatti, Narosa Series in Power and Energy Systems, Narosa Publishing House.
3. Variable Speed Generators, Ion Boldea, CRC press, Taylor & Francis Group.
4. Modeling and Control of Fuel Cells: Distributed Generation Applications, M. Hashem Nehrir and Caisheng Wang, IEEE Press Series on Power Engineering, Wiley-Interscience, A John Wiley & Sons, Inc, Publication.
5. Renewable and Efficient Electric Power Systems, Gilbert. M. Masters, Wiley-Interscience, A John Wiley & Sons, Inc, Publication.
6. Solar Photovoltaics Fundamentals, Technologies and Applications By Chetan Singh Solanki, PHI Learning Private Limited.

PhD Thesis/Report:

1. Analysis, Modeling and Control of Doubly-Fed Induction Generators for Wind Turbines, Andreas Petersson

2. Dynamic Wind Turbine Models in Power System Simulation Tool DIgSILENT, Anca.D.Hansem, Florin Iov, Poul Sorenson, Nicolaos Cutululis, Clemens Jauch, Frede Blaabjerg, Riso National Laboratory, DTU,Denmark
3. Wind PowerModelingandImpactonPowerSystemDynamics, J G Slootweg

Journal papers:

- IEEE Transactions on power system
- Electric power systems research
- Int. Journal of electric power and energy systems

EE764 DISTRIBUTED GENERATION AND SMART GRID (PE-2)**2ND SEMESTER AND 1ST YEAR****M.TECH (Electrical Engineering)****A. Credit Hours:**

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr No.	Title of Unit	Min. no of hours
1	Introduction to distributed generation	05
2	System Studies with distributed generation	10
3	Smart Grid	04
4	Data Communication and Communication Technologies in Smart Grid	08
5	DC Distribution & the Smart Grid	10
6	Power Electronics and Energy Storage in Smart Grid	08

Total Hours (Theory) : 45

Total Hours (Lab) : 30

Total Hours: 75

C. Detailed Syllabus

- | | | | |
|----------|--|-----------------|---------------|
| 1 | Introduction to distributed generation | 05 Hours | 11.12% |
| | Reasons for distributed generation, DG Technologies, Issues of distributed generation and their Technical impacts on the distribution system | | |
| 2 | System Studies with distributed generation | 10 Hours | 22.22% |
| | Types of system studies, Power flow studies, fault studies, stability studies, protection issues with distributed generations | | |
| 3 | Smart Grid | 04 Hours | 8.88 |
| | Smart grid infrastructure, benefits, Enabling High Penetration of Clean, Renewable Energy, Integrating Electric Vehicles to the Grid | | |
| 4 | Data Communication and Communication Technologies in | 08 Hours | 17.78% |

Smart Grid

Dedicated and shared communication channels, Switching techniques, Communication channels, Layered architecture and protocols, Communication technologies, Standards for information exchange, Cyber security standards, Smart metering, Smart meters: An overview of the hardware used, Communications infrastructure and protocols for smart metering

5 DC Distribution & the Smart Grid 10 Hours 22.22%

Ac vs. Dc power: an historical perspective, centralization dictates ac instead of dc, benefits and drivers of dc power delivery systems, powering equipment and appliances with dc, data centers and information Technology (it) loads, potential future work and research, launching the intelligrid, the intelligrid today, a smart grid vision based onThe intelligrid architecture, barriers to achieving the vision of intelligrid

6 Power Electronics and Energy Storage in Smart Grid 08 Hours 17.78%

CSC and VSC for low and medium power applications, Energy storage technologies, FACTS, HVDC, active filtering, D-STATCOM

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of distributed generations like solar, wind etc and incorporate them in the conventional power system problem analysis. .
CO2	To create and apply appropriate solution methods to solve smart grid problems.
CO3	Apply latest Artificial intelligence methods to solve OPF problems of smart grid in the presence of RES and electric vehicles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	-	3	-	2
CO2	2	-	-	1
CO3	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lecturers will be conducted with aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory, which carries five marks in overall evaluation.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students for each unit and will be evaluated at regular interval. It carries a weightage of 5 marks in the overall internal evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials will be conducted having a share of 5 marks in the overall internal evaluation.
- At the end of the course, the students will acquire the knowledge about the operational aspects of distributed generation and smart grid.

F. Recommended Study Material:

Books:

1. Vehbi C. Güngör, Dilan Sahin, Taskin Kocak, Salih Ergüt, Concettina Buccella, Carlo Cecati, and Gerhard P. Hancke, Smart Grid Technologies: Communication Technologies and Standards IEEE Transactions On Industrial Informatics, Vol. 7, No. 4, November 2011.
2. Xi Fang, Satyajayant Misra, Guoliang Xue, and Dejun Yang “Smart Grid – The New and Improved Power Grid: A Survey” , IEEE Transaction on Smart Grids,
3. Stuart Borlase “Smart Grid Infrastructure, Technology and Solutions”, CRC Press 2012.
4. Janaka Ekanayake, Nick Jenkins, Kithsiri Liyanage, Jianzhong Wu, Akihiko Yokoyama, “Smart Grid: Technology and Applications”, Wiley.
5. Nick Jenkins, Ron Allan et. Al., Embedded Generation, IET Power and Energy Series

Web Materials

1. IEEE Transactions of Smart Grid
2. NPTEL Course on Power Electronics and Distributed Generation by Prof. Vinod John

3. J.A. Pecas Lopes “Integrating distributed generation into electric power systems:A review of drivers, challenges and opportunities”, Electrical Power System Research, 77 (2007), 1189–1203

EE772: RESTRUCTURING AND DEREGULATION OF POWER SYSTEM (CT-2)**2ND SEMESTER AND 1ST YEAR****M.TECH (Electrical Engineering)**

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of Unit	Hrs
1	Introduction to Deregulation in Power System	02
2	Power System Restructuring Model	03
3	Available Transfer Capability (ATC)	10
4	Congestion Management	10
5	Transmission Pricing	05
6	Ancillary Services	10
7	Optimal Bidding in Restructured Power System	05

Total hours (Theory) : 45

Total hours (Lab) : 30

Total hours :75

C. Detailed Syllabus:

1. Introduction to Deregulation in Power System 02 4.44%
Hours

Concepts of regulation and deregulation, Characteristic of regulated power system, Need to restructured the power system, Overview of a deregulated industry, Disaggregation of traditionally vertically integrated utility, Structure of deregulated power system, Different entities in deregulated power system, Responsibilities of independent system operator, Trading arrangements: Pool, Bilateral, Multilateral, Power Exchange, Energy auction and market clearing prices

2. Power System Restructuring Model 03 6.66%
Hours

Classification of market models, Models based on energy trading, Models based on contractual agreements, Models based on ISO, Market operations, Day-Ahead and Hour-ahead markets, Elastic and Inelastic market, Market power, Vertical and horizontal market power, Measuring market power

3. **Available Transfer Capability (ATC)** 10 Hours 22.22%
Definition of ATC, Criteria for ATC evaluation, Methods of ATC calculation, Power transfer distribution factor (PTDF) and Line outage distribution factor (LODF), Optimal power flow (OPF), Continuation power flow (CPF), Open Access Same-Time Information System (OASIS), Structure of OASIS, Functionality of OASIS, Posting of information on OASIS, Information requirement of OASIS
4. **Congestion Management** 10 Hours 22.22%
Definition and effects of congestion in restructured power system, General methodologies for congestion management, Transaction curtailment, Transmission capacity reservation, System re-dispatch, Overall congestion management process, Methods for congestion management, Available transfer capability (ATC), Price area congestion management, Optimal power flow (OPF), Formulation of intra-zonal and inter-zonal problem for congestion management, Market based dynamic congestion management
5. **Transmission Pricing** 05 11.11%
Hours
Cost component of transmission system, Pricing of transmission service, Transmission pricing methods.
6. **Ancillary services** 10 Hours 22.22%
Ancillary services in restructured power system, Global overview of ancillary service management, Reactive power as an ancillary services, Reactive power placement analysis, Nodal pricing of reactive power, Costing and pricing of third party reactive power support, Synchronous generator as an ancillary service provider, Frequency regulation as an ancillary service, Spinning reserve service, Black start capability service
7. **Optimal Bidding in Restructured Power System** 05 11.11%
Hours
Game theory – an overview, Framing of bidding problem in game theory, Example, Optimization based approach for making bidding, Optimization based market simulator, Market prices with GENCOs behavior, Markov decision process

D. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Interpret the fundamentals of Restructured power system and use them to solve various complex problems of restructured power system.
CO2	To create and apply appropriate solution methods to solve power system economics problems
CO3	Apply latest Artificial intelligence methods to solve bidding optimization of smart grid in the presence of RES and electric vehicles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	-	-	2
CO2	-	2	-	-
CO3	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lecturers will be conducted with aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory, which carries 10 marks in overall evaluation.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students for each unit and will be evaluated at regular interval. It carries a weightage of 5 marks in the overall internal evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lecturers.
- At the end of course, the students will acquire the knowledge regarding the benefits of restructured power system, its problems and remedial measures.

F. Recommended Study Material:

Books:

1. L. L. Lai, Power System Restructuring and Deregulation, John Wiley & Sons, UK, 2001.
2. M. Shahidehpour and M. Alomoush, Restructured Electrical Power Systems, Operation, Trading, and Volatility, Marcel Dekker, 2001.
3. K. Bhattacharya, M. H. J. Bollen and J. E. Daalder, Operation of Restructured Power Systems, Kluwer Academic Publishers, 2001.
4. Yong-Hua Song and Xi-Fan Wang (Eds), Operation of Market-oriented Power Systems, Springer, 2003.

EC770 DIGITAL IMAGE & VIDEO PROCESSING (CT-2)

M.TECH 2nd SEMESTER

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Fundamental of Digital Image Processing	3
2.	Image Transforms	7
3.	Image enhancement	6
4.	Image Restoration	6
5.	Image Compression	8
6.	Video formation and representation	8
7.	2D Motion Estimation	7

Total Hours (Theory):45

Total Hours (Lab): 30

Total Hours:75

C. Detailed Syllabus:

	This will provide details about topics under each units of the course.		
1.	Fundamental of Digital Image Processing	03 Hours	8 %
1.1	Fundamental steps in Digital Image Processing		
1.2	Sampling and quantization,		
1.3	Basic relationship between pixels, 2-D Convolution		
2.	Image Transforms	07 Hours	16 %
2.1	2-D Discrete Fourier Transform		
2.2	Discrete Cosine Transform		
2.3	Discrete Wavelet transform		
3.	Image Enhancement	06 Hours	13 %
3.1	Basic Gray Level Transformations, Histogram Processing		

3.2	Enhancement using Arithmetic /Logic operations		
3.3	Image Smoothing and sharpening in spatial domain		
3.4	Image Smoothing and sharpening in frequency domain		
3.5	Edge detection and thresholding		
4.	Image Restoration	06 Hours	13 %
4.1	Noise Models, Restoration in the Presence of Noise Only–Spatial Filtering		
4.2	Inverse Filtering ,Minimum Mean Square Error (Wiener) Filtering, Constrained Least Squares Filtering,		
4.3	Geometric Mean Filter, Geometric Transformations		
5.	Image Compression	08 Hours	17 %
5.1	coding Redundancy, spatial and temporal redundancy		
5.2	Compression models : Lossy and Lossless		
5.3	Huffmann coding, Arithmetic coding, LZW coding, run length coding, Bit Plane coding, transform coding, predictive coding ,		
5.4	wavelet coding,		
5.5	Introduction JPEG standards		
6.	Video formation and representation	08Hours	17 %
6.1	Digital Video, Time varying image formation model: 2D motion models		
6.2	Fourier analysis of video signals and frequency response of Human visual system		
6.3	sampling of video signals and sampling rate conversion		
6.4	Video modelling		
7.	2-D Motion Estimation	07 Hours	16 %
7.1	Optical flow, general methodologies,		
7.2	pixel based motion estimation		
7.3	Block matching algorithm		
7.4	Mesh based motion Estimation, global Motion Estimation,		
7.5	Application of motion estimation in video coding.		

D Instructional Methods and Pedagogy:

- Multimedia Projector
- OHP
- Audio Visual Presentations
- Chalk + Board
- White Board
- Online Demo

- Charts

E Student Learning Outcomes / objectives:

- Enhancement of various images
- Able to restore the images from noises
- Video formation from images
- Motion estimation algorithms
- Video coding introduction .

F Recommended Study Material:

Text Books:

1. S. Jayaraman, S. Esakkirajan & T. Veerakumar, "Digital Image Processing", Tata McGraw Hill Education Pvt. Ltd. Second Edition.
2. Yao wang, Joem Ostarmann and Ya – quin Zhang, "Video processing and communication ",1st edition , PHI

Reference books:

1. Digital Image Processing, Rafael Gonzalez, Richard Woods, 2nd ed., Pearson Education
2. Fundamentals of Digital Image Processing, A.K. Jain ,Prentice Hall of India
3. M. Tekalp , "Digital video Processing", Prentice Hall International

Websites:

www.qi.tnw.tudelft.nl/Courses/FIP/noframes/fip.html
<http://eeweb.poly.edu/~yao/EL6123old/>
www.imageprocessingplace.com/root_files.../tutorials.htm

**CE768: SOFTWARE PROJECT MANAGEMENT & QUALITY
ASSURANCE (CT-2)
M.TECH 2nd SEMESTER**

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the course:

Sr No.	Title of the unit	Minimum number of hours
1.	Introduction to Software Project Management System	06
2.	Project Tracking and Configuration Management	08
3.	Introduction to Software Quality Assurance	05
4.	Software quality Assurance (SQA) Management and Software Quality Metrics	08
5.	Software Quality Engineering and Inspection and Defect prevention	06
6.	Software Testing and Maintenance	08
7.	Software Testing Tools	04

Total hours (Theory): 45Hrs.

Total hours (Lab): 30Hrs.

Total hours: 75Hrs.

B. Detailed Syllabus:

- 1. Introduction to Software Project Management System** 06 Hours 13 %
Overview of Project Planning, Project Estimation, Project Scheduling, Organization and Team Structure, Risk Analysis and Management, Resource Allocation
- 2. Project Tracking and Configuration Management** 08 Hours 17 %
Measurement of Physical and Financial progress, Earned value analysis, Status reports and Milestone reports, SCM activities, Standards for Configuration Audit Functions, Personnel in SCM Activities, Change control, Source code Control System (SCCS), Software Configuration Management: Some Pitfalls
- 3. Introduction to Software Quality Assurance** 05 Hours 11 %
Quality Control, Assurance, Movements, SQA-Software Quality Assurance Activities,

Approaches To SQA, Reliability, ISO 9000 And 9001, CMM Levels, Quality Audit, Concepts of Quality Improvement, Concepts of Process Maturity, Improving Process Maturity, IDEALSM Model for Process Improvement

4. Software quality Assurance (SQA) Management and Software Quality Metrics 08 Hours 17 %

Overview of SQA planning, techniques and contents of a SQA plan , establishing quality goals - Quality Function Deployment-Goal/Question/Measure Paradigm, total quality Management, cost of quality, quality assurance management, quality standards, factors affecting SQA effort, Management review process - technical review process -software assertion process - walkthrough process - audit process - verification & validation, Measuring quality, measurement criteria, product and process quality metrics, metrics for configuration management and software maintenance, example of metrics programs, complexity metrics and their relationship with testing and quality, metrics for object-oriented software analysis.

5. Software Quality Engineering, Inspection and Defect prevention 06 Hours 13 %

Defining Quality Requirements, Complexity Metrics and models, Project Tracking and Oversight, Data Quality Control, Software Inspection, Reliability Models, Reliability Growth Models

6. Software Testing and Maintenance 08 Hours 17 %

Foundations of Testing, Test Planning, Test Design and Implementation, Testing Network Management Systems, Web Based Testing, Testing Object-Oriented systems, Test Execution and Measurement, Management Issues for Software Quality, Software Testing Types: Unit, Integration, & System, Benchmarking and Certification, Control flow & loop testing, Data-flow testing, Transaction-flow testing, Domain testing, Coverage vs. usage based testing, Software Reuse, Software Aging, Product Enhancement, Reverse Engineering, Re-engineering Method, Architectural Simplification

7. Software Testing Tools 04 Hours 12 %

Test case Generation Methodology, Study of various Testing Tools (Win Runner, Load Runner), Automatic Testing Tool.

C. Instructional Method and Pedagogy:

- Lectures will be taken in class room with the use of multi-media presentations and black board – mix of both.

- Assignments based on above course content will be given to the students at the end of each chapter. Each assignment contains minimum 5 questions.
- Quizzes and Surprise tests will be conducted for testing the knowledge of students for particular topic.

E. Student Learning Outcome:

Upon successful completion of this course, students will be able to understand software project management process and different aspect of development process necessary for the management of the project which includes various activities, resources, quality, cost and system configuration etc. Student will learn the issues related to design and development of good quality software, data gathering, and interpretation and learn the relevant techniques and quality models. Students will study the various topics relevant to Software Quality and Testing. It focuses on types of testing and test case generation for testing the software. It introduces the testing tool to test the system. Student will also learn defect prevention and software maintenance.

F. Recommended Study Material:

❖ Text Books:

1. Pankaj Jalote, "Software Project Management in Practice", 2002, Pearson, Education Asia.
2. Roger S. Pressman, "Software Engineering: A practical Approach", Fifth Edition 2001, McGraw-Hill.
3. Bob Hughes and Mike Cotterell, "Software Project Management", Third Edition 2002, McGraw-Hill.

❖ Reference Books:

1. Rapid Testing" by Robert Culbertson, Chris Brown and Gary Cobb; Prentice-Hall, 2002.ISBN 0-13-091294-8
2. Metrics and Models in Software Quality by Stephen Kan, Addison-Wesley.
3. Software Engineering By Ian Sommerville Addison Wesley
4. Fundamentals of Software Engineering By Rajib Mall, Prentice Hall of India

5. The Capability Maturity Model: Guidelines for Improving the Software Process by Mark Paulik, Addison-Wesley.
6. "Black-Box Testing: Techniques for Functional Testing of Software and Systems", by Boris Beizer, John Wiley & Sons, Inc., 1995. ISBN# 0-471-12094

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS705 (A/B/C/E/F/G/H): ACADEMIC WRITING

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS705	Academic Writing	02	02	–	–	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	Academic Writing and Research Process • Introduction to Academic Writing • Academic Writing as a Part of Research • Types of Academic Writing • Features of Academic Writing • Importance of Good Academic Writing in various Academic Works	05
2	Anatomy of Academic Writing • Academic Vocabulary • Simple and Complex Sentences • Organizing Paragraphs • The Writing Process • Adopting Academic Writing Style	05
3	Key Academic Skills • Note – taking • Note – making • Paraphrasing • Summarizing	05
4	Accuracy in Academic Writing • Lexical Range • Academic Language and Structures • Elements of Writing • Proof Reading, Editing, and Rewriting	05
5	Using and Citing Sources of Ideas • Academic Texts and their Types • Intellectual Honesty in Academic Writing • Avoiding Plagiarism – Idea Theft • Degrees of Plagiarism	05

	• <i>Types of Borrowing</i> • <i>Anatomy of Citations</i> • <i>Common Citation Styles</i>	
6	Contemporary Practices in Academic Writing • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	05
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
6	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical / Quiz / Project / Academic Writing	–	70	70
Total				70

V. Course Outcomes (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
C05	understand the concept of plagiarism and learn to use citation and referencing styles

Course Articulation Matrix:

	PO1	PO2	PO3
CO1	1	3	1
CO2	-	3	-
CO3	-	3	-
CO4	-	3	-
C05	-	2	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Reference Books / Reading

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- *Effective Academic Writing Level - 1,2,3,4 (Second Edition)* By: Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; Publisher: Oxford

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

M. Tech. (EE) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE811: PROJECT PRELIMINARIES
M TECH 3rd SEM ELECTRICAL POWER SYSTEM

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	4
Marks	-	-	200	

A Outline of the Course:

- Project Preliminaries include course work on a specialized Subject or a Seminar.
- The course work shall be related to the area of his/her project research work.
- The coursework may be chosen from the existing PG (M. Tech.) Programmes of the registering department or from those of other departments.
- In addition to existing courses of the M. Tech. Programme, a department may offer special courses or seminar topics to the students.
- The specific subject of course work study will be decided by the department level Post Graduate Committee on recommendation of the supervisor(s).
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Resident students will satisfy the course requirement by attending Institute and Departmental courses, and submitting technical writings on assigned course.
- Non-resident students (full time research in R&D institutions/industries) will submit equivalent amount of technical writing and reports on seminars from their own places of work.
- Course offered at 3rd Semester will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- The course work and report is expected to show clarity of thought and expression, critical appreciation of existing literature and analytical or experiment or software based applications in the field of specialization.

- Course work can be based on topic of the Project Work. It may include literature review, required theoretical input, study and comparison of various approaches for the proposed project work.
- A student has to produce some useful outcome by conducting experiments or simulations.
- Student can learn all aspects and functionality of specialized software in the institute.
- Student can generate one working test bench using research paper along with results verified from referred paper through simulation and by modify existing work student can produce one research paper (research studies)

B Instructional Methods and Pedagogy:

- Students will select related topic based on subjects they learnt and other literatures like books, periodicals, journals and various internet resources.
- Students can select the topic based on the research areas of available supervisor/guide.
- Each student has to prepare a write-up of about 30-40 pages in prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval by the guide and endorsement of the Head of Department.
- Performance of students at the seminar will be assessed by the department level Post Graduate Committee. The supervisor/guide based on the quality of work and preparation and understanding of the candidate shall do an assessment of the course.

C Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	At the end of the course the student's gets exposure to design a research investigation that incorporates appropriate theoretical approaches, conceptual models, and a review of the existing literature.
CO2	Students will explore the new ideas and the possible areas to work ahead.
CO3	Student will learn the various research methodologies useful for doing project work.
CO4	Student will learn to investigate the chosen topic in depth. This implies collecting and reviewing literature and understanding and interpreting the most up-to-date concepts and theories of your chosen academic field and/or project topic.
CO5	Student will learn to apply the concepts and theories learnt in previous years of study and work placements.

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	2	3	2
CO2	3	2	1	2
CO3	2	3	3	2
CO4	3	3	2	2
CO5	3	2	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

1. Programming Languages: C, C⁺⁺, MATLAB
2. Softwares: MiPower, Power Factory, ANSYS, PSpice, SKM Power Tools, PS CAD, Dspace, CYME, EMPT-RV, Power World Simulator.
3. IEEE : www.ieeeexplore.ieee.org
4. www.sciencedirect.com
5. Indian Jr. of Engg. and Material Sc. (www.niscair.res.in)
6. Sadhna (Engineering Science) (<http://www.ias.ac.in/sadhana/>)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE812: PROJECT PHASE I
M TECH 3rd SEM, ELECTRICAL POWER SYSTEM

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	18
Marks	-	-	500	

B Outline of the Course:

- The Project shall be related to the major field of his/her PG specialization work.
- The Project should be one of the major pieces of evidence that students are familiar with or that student wants to be familiar with. It should reflect specialist subject by means of deep and sustained study.
- The project will be finalized by the department level Post Graduate Committee on recommendation of the supervisor(s).
- The project work shall be carried out by each candidate independently during the third and fourth semester under the guidance of one of the faculty members of the Department. If the project work is of inter-disciplinary nature, a co-guide shall be taken from the same or any other relevant Department.
- If a project work has to be carried out in any industry / factory / organization, outside the campus, the permission to that effect and the name of co-guide at any of these organizations shall be intimated to the Post Graduate Committee at the beginning of third semester.
- Project Phase I includes literature review, required theoretical input, study and comparison of various approaches for the proposed dissertation work.

C Instructional Methods and Pedagogy:

- Student has to submit a project/dissertation proposal indicating the tentative title and broad outline of the proposed work and the name(s) of the supervisor(s) along-with their concurrence in writing within 30 days from the starting of the third semester.

- Utmost care should be taken in selection of research topic so that repetition of research work is avoided.
- Project Phase- I will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- After successful completion of Project Phase I only students are allowed to go register for Project Phase - II.

D Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	Student will judge a topic that is appropriate for his/her degree specialization.
CO2	At the end of the course the student's gets exposure to construct and justify research questions related to the topic.
CO3	Student will be in a position to design a research investigation that incorporates appropriate theoretical approaches, conceptual models, and a review of the existing literature.
CO4	Student will learn to structure a discussion in a coherent and convincing way by synthesizing the material in the context of the research questions.
CO5	Student will be having sufficient collection of the literature/experimental data for the implantation/experimentation in Project Phase - II.

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	2	3	2
CO2	3	2	1	2
CO3	2	3	3	2
CO4	3	3	2	2
CO5	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

- 1 Programming Languages: C, C⁺⁺, MATLAB
- 2 Softwares: MiPower, Power Factory, ANSYS, PSpice, SKM Power Tools, PS CAD, Dspace, CYME, EMPT-RV, Power World Simulator.
- 3 IEEE : www.ieeeexplore.ieee.org
- 4 www.sciencedirect.com
- 5 Indian Jr. of Engg. and Material Sc. (www.niscair.res.in)
- 6 Sadhna (Engineering Science) (<http://www.ias.ac.in/sadhana/>)

M. Tech. (EE) Programme

SYLLABI (Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING
EE813: PROJECT PHASE II
M TECH 4th SEM ELECTRICAL POWER SYSTEM

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	-	-	30
Marks	-	-	1000	

A Outline of the Course:

- Student should carry out the investigation by identifying sources of evidence, accessing those using accepted and rigorous academic methods, and analysing and interpreting the material gathered by simulation/experimentation.
- A Project Phase- II is student's own work and will need to keep up the effort, and the interest, over several months and through several stages.
- Student need to think carefully about the time necessary to carry-out and complete your project work and the relative writing up.
- The project should present an orderly and critical exposition of the existing knowledge of the subject and will embody results of original investigations demonstrating the capacity of the candidate to do independent research work.
- While writing the thesis/dissertation, the candidate will layout clearly the work done by him independently and the sources from which he has obtained other information contained in his/her Dissertation

B Instructional Methods and Pedagogy:

- Project Phase- II will be evaluated at least once during the semester and at the end of the semester as a part of continuous evaluation.
- Before submission of Phase II project/dissertation report, it is expected from a student to publish at least one research paper in National/International conference. Further, for such publications, Department Post Graduate Committee will identify and approve the

national/international conferences.

- The dissertation shall be submitted for ‘dissertation – evaluation’ ordinarily at the end of IV Semester and ‘dissertation – open defence shall be held soon after the submission of the dissertation.

C Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1	At the end of the course the student’s gets exposure to design a research investigation that incorporates appropriate theoretical approaches, conceptual models, and a review of the existing literature.
CO2	Students will learn to structure a discussion in a coherent and convincing way by summarizing the key arguments and providing suitable and coherent findings.
CO3	Student will be able to draw valid conclusions, relating them to the research topic.
CO4	Students will write a comprehensive review of the literature, including a review of other dissertation research related to their study.
CO5	Students develop a design of their study with a discussion of the methodology to be used including selection of a sample, instrumentation and its testing, sources of data and the data collection process.
CO6	Students describe how their data will be treated and analysed and the significance and limitations of their study.

Course Articulation Matrix:

Course Outcomes	PO1	PO2	PO3	PSO1
CO1	3	2	3	2
CO2	3	2	1	2
CO3	2	3	3	2
CO4	3	3	2	3
CO5	3	2	3	3

CO6	3	3	3	3
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1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D Recommended Study Material:

- 1 Programming Languages: C, C⁺⁺, MATLAB
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